

1 PROTOCOL FOR:

2 A fast and simple fluorometric method to detect cell death in 3D intestinal organoids

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19
20 LEGEND

21  **ATTENTION**

22 * **HINT**

23  **REST**

24 REAGENTS AND MATERIALS

25 Reagents

Chemicals/solutions	Abbreviation/Synonym	Catalog Number	Manufacturer
Propidium Iodide 1mg/ml solution	PI	P3566	Fisher Scientific
Hoechst 33342, Trihydrochloride, Trihydrate	Hoechst	H1399	Fisher Scientific
<i>cis</i> -Diammineplatinum(II) dichloride	Cisplatin	P4394	Sigma
5-Fluorouracil	5-FU	F6627	Sigma
Staurosporine		ALX-380-014-C250	Enzo Life Sciences
mEGF		315-09	Peprtech
mNoggin		250-38	Peprtech
hR-Spondin 1		120-38	Peprtech
Cultrex® Reduced Growth Factor Basement Membrane Matrix, Type 2 (BME 2)	BME	3533-010-02	Trevigen
BD Matrigel Basal membran matrix	Matrigel	734-0269	VWR
N2 supplement	N2	17502-048	Fisher Scientific
B27 supplement	B27	17504-044	Fisher Scientific

Dulbecco's Modified Eagle's Medium F12	DMEM-F12	D6421	Sigma
Dulbecco's Modified Eagle's Medium - high glucose; Phenol red free	DMEM – phenol red free	D1145	Sigma
Penicillin-Streptomycin	PenStrep	P4333	Sigma
Gentamicin solution	Gentamicin	G1272	Sigma
L-Glutamine	Glutamine	G7513	Sigma
Phosphate buffered saline	PBS	D8537	Sigma
HEPES		9105.3	Roth
N-acetyl-cysteine	NAC	A9165	Sigma
3-(4,5-Dimethylthiazol-2-yl)-2,5-diphenyltetrazoliumbromid	MTT	M2128	Sigma
Bovine Serum Albumine	BSA	T844.3	Roth

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27 Materials

Cell culture ware	Abbreviation/Synonym	Catalog Number	Manufacturer
5ml pipet		861.253.001	Sarstedt
10ml pipet		861.254.001	Sarstedt
25ml pipet		861.685.001	Sarstedt

10 µl pipet tip		70.760.002	Sarstedt
200 µl pipet tip		70.1130	Sarstedt
1000 µl pipet tip		70.762.411	Sarstedt
15ml Falcon Tube	15ml Falcon	62.554.502	Sarstedt
50ml Falcon Tube	50ml Falcon	62.547.004	Sarstedt
24-well plate	24-well plate	83.3922	Sarstedt
96-well plate	96-well plate	83.3924	Sarstedt

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29 PROCEDURE

30 Staining of organoids with Propidium Iodide & Hoechst33342

- 31 • Grow murine intestinal organoids as described in [1], preferably in clear 96-well plates.
- 32 • Plate 200-300 crypts per well in 8 µl Matrigel/BME and cover with 80 µl pre-warmed
- 33 complete murine intestinal organoid growth medium.
- 34 • Depending on whether starting with freshly isolated crypts or thawed organoids,
- 35 determine timepoint/duration of treatment and analysis.
- 36 •  Inspect organoid growth and integrity microscopically before starting treatment.

- 37 •  Treat organoids with substances of interest – use technical triplicates at least.
- 38 •  Include negative (untreated) and positive (Staurosporine) control.
- 39 Treat for at least 16 h to ensure disruption of plasma membranes.
- 40 •  Check effects of treatment – morphological changes in treated vs. untreated organoids
- 41 • Remove treatment medium and replenish organoids with pre-warmed staining medium
- 42 with $c(\text{final}) = 10\mu\text{g/ml}$ Propidium Iodide (PI) & Hoechst33342 (Hoechst).
- 43 •  Stain for 30 minutes at 37°C , 5% CO_2 .
- 44 • Replace staining medium with 80 μl pre-warmed fresh phenol red-free medium.
- 45 •  Check fluorescence with microscope.
- 46 • Start quantification.

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48 Quantification of Propidium Iodide & Hoechst33342 fluorescence

- 49 • Start plate reader and measurement software (Tecan).
- 50 •  Set measurement from top position → remove lid of the plate. Measurement can be
- 51 performed at room temperature, as it is an endpoint assay.

-  First, the Gain must be set to the wells for the highest expected cell death (PI) and the lowest expected cell death (Hoechst) and must be checked for values between 115 and 160 (positive and negative control).
-  Then, let the plate reader determine the optimal Z-position automatically from the corresponding wells. Check for values between 1.5×10^6 and $1.6 \times 10^6 \mu\text{m}$. (This can take up to 2 minutes)
- Measure fluorescence with 25 flashes, with an integration time of 20 μs . Lag and settle time are set to 0s.
-  Use four times four measurements per well, a boarder of 1mm around measurement points (measurement points can be increased to counter uneven organoid distribution, however on the cost of measurement speed).
- Excitation and emission wavelengths for PI are 535 nm and 617 nm, respectively. For Hoechst, 361 nm and 486 nm should be used as excitation and emission wavelengths.
- First, measure all wells for PI fluorescence and after a 30 s wait timer for Hoechst fluorescence.
-  Check first measurement points for data in range → if out of range, reduce Gain and

re-measure.

- First, calculate the PI/H ratio by using the relative fluorescence units (RFUs) of PI & Hoechst:

$$\frac{PI}{H} ratio = \frac{RFU (PI)}{RFU (Hoechst)}$$

- Then, use PI/H ratios to quantify treatment specific organoid cell death, by dividing each sample with the positive control Staurosporine (STS), multiplying it x100 and subtracting the PI/H ratio of the untreated (ut) samples:

$$treatment\ specific\ organoid\ cell\ death\ [\%] = \left(\frac{x(sample)}{z(STS)} \right) * 100 - y(ut)$$

RECIPES

Murine intestinal organoid growth medium

Murine small intestinal crypts were isolated and grown as described in [1, 2]. Growth medium for murine intestinal organoids was prepared as described in [2, 3], with minor modifications. In short, DMEM/F12 (Sigma), was supplemented with 0.1 % BSA (PAA), 2 mM L-glutamine

(Sigma), 10 mM HEPES (Sigma), 100 U/ml penicillin (Sigma), 100 μ g/ml streptomycin (Sigma), 1 mM N-acetyl cysteine (Sigma), 1 $\times$ B27 supplement (Invitrogen), 1 $\times$ N2 supplement (Invitrogen), 50 ng/ml mEGF (Peprotech), 100 ng/ml mNoggin (Peprotech).
Staining of intestinal organoids with MTT and measuring formazan absorption to quantify organoid cell death was described in detail in [2, 3].

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90 Propidium Iodide and Hoechst33342 staining medium

91 Use fully supplemented medium (as described above) and add Propidium Iodide and
92 Hoechst33342 up to a final concentration of 10 μ g/ml PI& Hoechst. Add 80 μ l of staining
93 medium per well (96-well plate).

94 EQUIPMENT

Equipment	Abbreviation/Synonym	Catalog Number	Manufacturer
Tecan Infinite M200 Pro – Platereader	Tecan		Tecan
Axio Observer.Z1	Fluorescence Microscope		Zeiss
HeraCell CO ₂ /O ₂ -Incubator	Incubator		Fisher Scientific

HeraSafe Laminar Flow			Fisher Scientific

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96 TROUBLESHOOTING

- 97 • Ensure organoid and Matrigel/BME integrity before starting treatment.
- 98 • Determine optimal time-point for start and endpoint of treatment (kinetics).
- 99 • Always add growth and staining medium gently to not disrupt Matrigel/BME domes.
- 100 • Check PI& Hoechst fluorescence microscopically before starting quantification.
- 101 • If effects of treatment are low, consider using thawed organoids rather than organoids
- 102 derived from freshly isolated crypts, in order to reduce unspecific background (dead cells
- 103 and cellular debris).

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105 References

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